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Sailing Around the Carbon Tax: Vessel Behaviour Under the EU Emissions Trading System

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Abstract

This paper provides an empirical evaluation of how the shipping fleet responded to the extension of the EU Emissions Trading System (EU ETS) that came into force in January 2024. Using a difference in difference analysis on a panel of 15,781 bulk vessels observed between 2013 to 2025, we measure the impact of this policy by comparing their behavioral responses based on how much of their trade was taxed. Because the tax is only applied on trips occurring within European waters, vessels face different cost levels depending on their usual trading routes. To capture this heterogeneity, we implement a Principal Component Analysis (PCA) and hierarchical clustering to split the fleet into three distinct strategic profiles based on their trade exposure to EU (Low, Medium, and High EU trade intensity). Globally, when we implement our main specification, that is a trend controlled model that includes an interaction between exposure and a continuous time index, we find no significant effect of the policy on most outcome variables, apart for CO_2 emissions per nautical mile and voyage distance. However, we discover striking results when we run this trend controlled analysis by cluster type. Specifically, we observe a statistically significant decrease in speed for the ships among every clusters, with the sharpest speed reduction centered on the intermediate exposure group. This effect was associated with shorter voyage distance (approximately 413 nautical miles per month) for the vessels more focus on EU trade, whereas the least exposed fleet expanded theirs by 773 nautical miles, providing a clear sign of carbon leakage through rerouting away from European ports. In addition to changes in the trading routes, the policy also induced a switch in the type of fuel used within the high exposure cluster, where standard fuel (VLSFO) was substituted for a cleaner option (MGO). Yet, most paradoxically, the most heavily regulated vessels actually experienced a decrease in CO_2 efficiency per nautical mile, as shortening their routes forced them into more frequent port stops and fuel inefficient trips, which completely wiped out the benefits of slowing down.

1. Introduction

Maritime shipping carries more than 80% of global trade by volume (UNCTAD, 2023). The shipping industry remained for a long time largely exempt from carbon regulation. This is not a small gap. Indeed, international shipping accounts for roughly 2% of global CO_2 emissions, with bulk shipping accounting for around 23% of these emissions (IMO, 2020). Even though comparable in scale to the aviation sector, the shipping industry never faced as serious carbon regulations as its airborne counterpart. The latter indeed suffered the consequences of the 2012 EU ETS policy expansion, making it internalize the CO_2 emissions from all flights within the European Economic Area (EEA). For a sector that carries most of what the world consumes, that gap is hard to ignore.

The economic logic behind it is straightforward. When a cargo owner orders a ship to move iron ore from Brazil to Rotterdam for instance, the carbon cost of that voyage is not born by any of the economic agents involved. Shipping firms internalize fuel costs, port fees, and crew wages, but greenhouse gas emissions fall on the agents outside the transaction. This is a textbook

negative externality in the sense of Pigou (1920): the market equilibrium produces too much shipping, and too many emissions, relative to what would be socially optimal. The standard fix would require pricing the externality, either through a carbon tax or through a cap and trade system.

That is exactly what the European Union did. In January 2024, the EU extended its Emissions Trading System to maritime shipping (European Commission, 2023). Shipowners now pay one EU Allowance (EUA) per tonne of CO_2 emitted on voyages involving European waters. In 2024, EUA prices ran between 55 and 75 euros per tonne.

But the policy has an unusual feature. It does not treat all voyages equally. Emissions on routes between two EU ports are covered at 100%. Emissions on routes between an EU and a non EU port are covered at only 50%. This asymmetry is the heart of the paper. A vessel can cut its compliance bill in half simply by stopping at a non EU port on the way. It does not need new technology or to slow down. It just needs to change its route. Whether this is actually happening, and what it means for global emissions, is what we study.

The policy's implementation has been rolled out gradually. By law, shipowners were required to cover 40% of their emissions in 2024, 70% in 2025, and ultimately 100% from 2026 onward. This gradual implementation reflects the slow pace of the shipping industry. Commercial vessels typically have an operational lifespan of 25 to 30 years (UNCTAD, 2023), and renegotiating freight contracts takes some time. Because we are studying the policy at the 40% intensity threshold, the slow dynamics of the shipping industry impacts our analysis . Our estimates should therefore be thought of as a lower bound on the behavioral responses that a full carbon price would generate. Examining how ships react once the 100% compliance obligation began in January 2026 is a natural next step once sufficient data becomes available.

We study the impact of this policy on a range of outcome variables potentially capturing the shipping responses to a carbon price hike, namely through sailing speed, voyage distance, fuel mix (Marine Gas Oil (MGO), High Sulphur Fuel Oil (HSFO) and Very Low Sulphur Fuel Oil (VLSFO)), as well as CO_2 emissions intensity per nautical mile. We use a panel of 15,781 bulk cargo vessels observed monthly from January 2013 to June 2025, containing roughly 1.76 million vessel observations. The key feature of our setting is that treatment is continuous rather than binary. Because the EU ETS applies to any vessel stopping at an EEA port regardless of their country of registration, there is no clean control group of entirely untreated ships. Instead, vessels differ in the extent to which their operations are covered by the regulation, as captured by our *ETS_Rate* variable: a monthly average of regulatory coverage fractions running from 0 for ships with no EU exposure to 1 for ships operating exclusively within the EEA. We combine this continuous treatment intensity with the January 2024 implementation date in a difference in differences (DiD) design.

Average effects can hide very different behaviors across vessel types. To deal with this, we use

Principal Component Analysis (PCA) and Ward hierarchical clustering on pre-reform voyage data to split the fleet into three trade profiles: *Low*, *Medium*, and *High EU* exposure. We then run our main specification separately within each cluster. As we expected, this decomposition is essential: the global average hides responses of opposite sign that only become visible once you separate the fleet.

Our main findings are the following. Firstly, at the global level and after controlling for vessel pre trends, only two outcomes show a robust response to the policy: CO_2 intensity per nautical mile and voyage distances increase slightly. Once we split the fleet by trade cluster, a more stunning picture emerges. All three groups reduced average speed significantly, with the sharpest decline being noticed in the medium EU cluster. Voyage pattern change is consistent with leakage. Indeed, the 413nm per month shortening of *High EU* vessels' voyages while the *Low EU* ones expanded theirs by 773nm could suggest a rerouting away from European ports onto non EEA routes. Vessels with *High EU* exposure also reallocated approximately 0.37 percentage points of their fuel mix away from standard VLSFO toward the comparatively cleaner MGO, a finding that merits particular attention. The most paradoxical finding concerns the CO_2 intensity result. The latter contradicts the true purpose of the EU ETS policy: the most heavily regulated vessels became less efficient per nautical mile. Shortening route distances forces them into shorter, port intensive trips that are structurally more emissions intensive than the extended voyages they replaced.

Taken together, the results suggest that the EU ETS is slowly beginning to influence how ships operate. Its design may however be redistributing emissions globally rather than reducing them. We discuss the policy implications in Section 6.

The remainder of the paper is structured as follows. Section 2 reviews the relevant literature. Section 3 describes the data and institutional background. Section 4 lays out the empirical strategy. Section 5 presents the main results. Finally, section 6 discusses policy implications, and section 7 concludes.

2. Literature review

Carbon pricing in shipping is new. Indeed, the policy only entered into force in January 2024, and the empirical evidence is still thin. In this section, we synthesize what is known from adjacent literatures to build our expectations and our empirical design.

The review is organized around five different topics: how the maritime ETS works, what carbon pricing does to transport behavior, how ships respond through speed and fuel, how companies reroute to avoid costs, and what econometric problems arise. We close with why dry bulk is a particularly clean setting for this question.

2.1. How the EU ETS shipping rules work and fit with other regulations

The shipping sector was exempt from major carbon regulation for a long time. That changed in January 2024, when Directive (EU) 2023/959 brought the sector into the EU ETS. Such a policy did not come out of nowhere. Regulation (EU) 2015/757 forced ships above 5,000 gross tonnage to track and report their emissions through the THETIS-MRV system from 2018 onward (Psaraftis et al., 2019).

In the same paper Psaraftis et al. (2019) actually look at the first year of MRV data and find that the regulation's geographic scope creates an important bias in its efficiency measures, making them hard to use in practice. Hence the takeaway of their analysis is simple: reporting requirements alone do not change how ships operate. Without a financial cost attached, there is no reason to behave differently.

The ETS itself phases in gradually. Ships above 5,000 GT pay for 100% of emissions on voyages between EU ports and 50% on voyages involving a non EU port. The obligation covered 40% of emissions in 2024 and did reach 100% in the begining of 2026. It is also worth to note that methane and nitrous oxide were added to the scope that same year (Kotzampasakis, 2025).

A second regulation, namely FuelEU Maritime (Reg. (EU) 2023/1805), runs alongside the ETS. Rather than pricing emissions, it sets a limit on the carbon intensity of the fuel itself, starting with a 2% reduction in 2025 and increasing the treshold to reach 80% by 2050. Lehmann et al. (2025) in their study simulate both regulations on a fleet of sixteen container ships and find that at current carbon prices, fuel switching is not yet profitable. Some operators may simply find it cheaper to pay the FuelEU penalty than to switch fuels.

On a more global point of view, the IMO revised its decarbonization targets in 2023, targeting a net zero around 2050 with a 20% cut by 2030. Its main instrument is the Carbon Intensity Indicator (CII), a rating system rather than a price. However there are still some issues with the CII; Shi et al. (2026) point out three problems with it: it inadvertently rewards ships sailing empty, punishes delays caused by port congestion, and favors long ocean routes over short regional routes. Fortunately for EU lawmakers, a price instrument such as the EU ETS does not share these distortions, which is part of why it is a more interesting object of study.

2.2. Carbon pricing in transport: what the broader evidence says

According to a wide body of literature, among transport sectors, emissions do not respond much to carbon charges. In their analysis, Rafaty et al. (2025) found out that the introduction of a carbon price reduces the growth rate of total CO_2 emissions by about 0.06% for every additional dollar per tonne of CO_2, compared to roughly 0.16% in the manufacturing sector. Interestingly, their estimates for road transport highlight an even smaller response to the carbon price level. We strongly believe that shipping is unlikely to be much different, and that is what we will prove in section 5.

Känzig (2025) studies carbon pricing in Europe and he found that while it does raise energy prices and stimulate green investment, it also slows economic activity and hurts lower income households more than others. He also noticed that the tough job market alone explain two thirds of the drop in consumer spending. Concerning shipping, we could deduce two implication following this study. First, emissions will not move much unless carbon prices get high enough. Second, keeping the policy politically viable depends on what happens to the tax revenues.

More closely related to our subject, concerning maritime transport specifically, Mundaca and Strand (2020) estimate price sensitivities ranging from −0.03 to −0.52 depending on the cargo type. The noticed that biggest responses are in heavy, low value bulk goods like iron ore and coal. According to those two researchers, a global 40 dollar carbon tax would cut shipping emissions by around 7%, but mostly through less trade rather than cleaner fuels.

That last point matters. Clean shipping technology, ammonia, methanol, synthetic fuels, is still far too expensive for most operators. Seyhan et al. (2026) shows this clearly for bulk carriers: bio fuels and bio LNG both generate financial losses at carbon prices between 70 and 100 euros per tonne. Even dual fuel LNG, which reduces CO_2 by around 27%, becomes a losing option once the EU begins charging for methane emissions in 2026. In the short run, operators have little choice but to adjust how they sail rather than what they burn.

2.3. How ships respond: speed and fuel

To get a clear overview on the matter, we should dive into basic physics first. The fuel a ship burns per nautical mile goes up with the cube of its speed. Slowing down is cheap and immediate. It should be the first thing shipowners do when carbon costs rise. Whether they actually do it is less obvious.

For instance, there is a study from Adland and Jia (2018) that looks at large capesize bulk carriers and fond that speeds do not react consistently to freight rates or fuel prices. What matters more is whether the ship is full or empty, its design speed, and how the contract is written. On one hand, under voyage charters, legal clauses force the ship to sail without delay. On the other hand, under time charters, the charterer pays the fuel bill and decides the speed. A carbon price alone cannot be expected to cut through these constraints.

However, the picture is cleaner for shorter routes. Ollila et al. (2025) did look at over 110,000 trips in Northern Europe and find a speed to price elasticity of around −0.08. It seems that the gap with Adland and Jia (2018) comes down to the type of ships and consequently the type of voyage. Indeed, short sea operators have room to adjust while deep sea bulk carriers, that are locked into rigid contracts, do not.

We noticed that fuel switching has mostly been studied through sulphur rules rather than carbon taxes. For instance, Klotz and Berazneva (2022) use 250 million satellite observations to study California's 2009 low sulphur fuel rule. The hips followed the law inside the regulated zone,

but adjusted by taking longer routes outside it to keep burning cheaper fuel. Distance inside the zone fell by 24 to 44% and speeds dropped by 8 to 13%; the pollution shifted rather than disappeared. Hansen-Lewis and Marcus (2025) find the same pattern for the North American emissions control zone: only 53% of the expected cuts actually happened. The EU ETS does the same, but for carbon instead of sulfur.

2.4. Rerouting and leakage

When a carbon price only covers part of the world, industries find ways around it. Concerning the manufacturing sector, early EU ETS phases showed little relocation because free permits kept costs down. But as the prices rose, Naegele and Zaklan (2019) fond clear evidence of leakage: European imports became more carbon intensive as domestic production got cleaner.

But the shipping sector is different from the manufacturing sector. You cannot move a factory, but a ship can simply take a different route. Dominioni et al. (2018) compare policy designs and argue that tracking cargo destination works better than tracking port stops, which is what EU lawmakers do. Wu et al. (2024) frame the problem as a prisoner's dilemma. Adding a non EU port stop to shorten the taxed leg makes sense for the individual shipowner, but raises global emissions since the detour burns extra fuel. This study finding is further strengthen by the analysis of Lagouvardou and Psaraftis (2022). Those two researchers show the same outcome for container ships: switching to North African hubs becomes profitable even at low carbon prices, once port fees and cargo handling are taken into account.

Whether this is already happening is debated. The European Commission's 2025 monitoring report blamed recent port traffic shifts on the Red Sea crisis. Transport & Environment (2026) mostly agree for 2024 but flag ambiguous patterns in the Eastern Mediterranean. Additionally, data from the EU ETS Observatory tells another story: the share of long distance container traffic at main EU ports fell from 67% in 2023 to 56% in 2025 while Egyptian ports grew from 35% to 52% of the Eastern Mediterranean market, a phenomena that we cannot attribute to the Red Sea crisis.

Kotzampasakis and Wooley (2025) in turn list four ways companies can dodge the tax: flag switching (which does not work here since the ETS applies regardless of port stop), adding no -EU port stops, putting cleaner ships on EU routes while dirtier ones sail elsewhere, and speeding up outside the taxed zone. The last two are the hardest to detect. Indeed, a ship's European emissions can fall while its global footprint quietly rises.

2.5. Econometric challenges

Our design uses a two way fixed effects panel model with a continuous treatment, a design that is detailed in section 4. There are a few econometric problems are worth noting before starting the analysis.

The first is serial correlation. Bertrand et al. (2004) show that ignoring it in a DiD designs produces false positives up to 45% of the time. The easiest fix is to cluster standard errors at the level where treatment is assigned, here at the vessel level, which is exactly what we have done. Cameron et al. (2011) extend this to settings where correlation runs in more than one direction.

The staggered DiD literature is also relevant is another interesting part of DiD design, even though our policy starts on a single date for every ships affected by it (January 2024). In a recent paper, Goodman-Bacon (2021) shows that standard TWFE estimates decompose into a weighted average of pairwise comparisons, some of which use already treated groups as controls. This biases results when effects change over time. Sun and Abraham (2021) find the same problem in event study designs. However, as explained earlier, staggered timing is not an issue here since all vessels face the same January 2024 start date. But the continuous treatment version of the problem is. Callaway et al. (2024) show that TWFE with a continuous dose can assign negative weights to some groups, making the coefficient hard to interpret without strong assumptions.

The parallel trends has also been challenged. Roth (2022) shows that testing for it before running the main model can introduce some bias. Rambachan and Roth (2023) propose a method that skips the pre-test entirely and builds confidence intervals that hold even when parallel trends is not perfect. Unfortunately, we cannot apply it here since the HonestDiD package only exists in R, and we are working in a Python environment.

Finally, Wolfers (2006) and Meer and West (2016) warn that adding unit specific time trends can erase the true policy effect when treatment changes the slope of behavior rather than causing an instantaneous shift. We deal with this by interacting treatment intensity with a continuous time index instead, which lets the model pick up the break at January 2024 without absorbing it.

The closest paper to ours in design is Economides (2025). He studies a port queuing policy at Los Angeles and Long Beach using vessel and time fixed effects with vessel clustered standard errors. He found a 17% drop in speeds and an 11.2% cut in emissions intensity. His result that the policy moved speeds more than fuel choices is a useful reference point for our own findings, and we shall find similar results.

2.6. Why dry bulk

We made the choice to focus on dry bulk rather than container ships for several reasons. First, the dry bulk market is close to perfect competition. There are many small owners, services are largely identical, and prices are set by global supply and demand rather than by forming a cartel. Therefore, any response to a new cost would shows up directly in the data.

The sector is also volatile. Greenwood and Hanson (2015) document return drops of 76% and surges of 86% across investment cycles, driven by owners all ordering new ships at the same time during strong markets. Kavussanos and Alizadeh-M (2001) find seasonal shifts of up to

18% within a single year for capesize vessels. This volatility is a problem we deal with using time fixed effects, but it also generates the kind of within ship variation that our identification needs to work properly.

The regulation affects ships differently according to their size. For instance Capesize ships above 100,000 DWT usually run fixed deep sea routes for iron ore and coal with almost no flexibility. Panamax vessels (60,000 to 100,000 DWT) in turn carry coal, grain, and bauxite on more varied routes. Smaller handysize and supramax vessels (which are from 10,000 to 60,000 DWT) carry their own loading gear and can stop at smaller ports. This kind of vessels have the most room to simply avoid European routes when the carbon cost gets too high. This tells us that the same carbon price can therefore produce very different responses depending on the ship type. That is exactly the heterogeneity our cluster analysis, described in 4 is built to capture.

Contract structure matters too. Under time charters the charterer pays the fuel bill and decides the speed, while the shipowner carries the long run compliance costs. In an interesting study, Adland and Jia (2018) show that this split weakens the link between fuel prices and speed choices. This same split likely limits the responses we find in our data. We therefore expect any effects to show up in the dimensions the charterer (for instance speed and routing), rather than in fuel technology or vessel upgrades.

Taken together, the literature gives us three baseline expectations. Speed should fall for vessels facing higher compliance costs. Similarly, routes should shorten for EU exposed ships and lengthen for the rest. Finally, fuel switching is possible but unlikely to be large at 2024 carbon prices level. These are the benchmarks we bring to the data.

3. Data and institutional background

3.1. Institutional background

The EU ETS maritime extension, entered into force on January 1, 2024, following a formal proposal by the European Commission in July 2021 and parliamentary adoption in April 2023. This 30 month gap between announcement and implementation is relevant from an econometric point of view at least: forward looking operators had ample time to begin adjusting behavior before the compliance date, meaning any pre-2024 changes detectable among EU-exposed vessels may reflect anticipation rather than a direct response to the carbon charge. All maritime sector permits are auctioned with no free allocation, and revenues are directed toward climate policy by member states.

A concurrent regulation, the IMO Carbon Intensity Indicator (CII), effective January 2023, assigns annual CO_2 intensity ratings to all vessels above 5,000 GT regardless of trade route. Because CII applies uniformly across all vessels, its monthly effect is common to the entire fleet and is fully absorbed by the calendar-month fixed effects in the estimator described in section 4.

CII does not confound the difference in differences estimates.

3.2. Data sources and sample construction

The primary dataset is a voyage level record of commercial maritime activity spanning January 2013 to June 2025, yielding approximately 3.8 million raw voyage observations. Each observation corresponds to a single voyage and records vessel name, departure date, departure country, commodity type, EU voyage classification, distance sailed (nautical miles), average speed (knots), fuel consumption by type (VLSFO, HSFO, and MGO, in tonnes), total CO_2 emissions (tonnes), leg type (laden or ballast), and vessel deadweight tonnage (DWT).

From these voyage level observations, a vessel month panel is constructed by grouping on (*vessel name*, *year-month*) and computing monthly means of all outcome and treatment variables. A vessel operating five voyages in a given month contributes a single row reflecting its average behavior that month. This aggregation smooths high frequency operational noise and produces the one observation per entity per period structure required by the two way fixed effects estimator. The raw 3.8 million voyages reduce to approximately 1.9 million vessel month observations before the balancing step.

The DiD design requires each vessel to be observed in both the pre reform and post reform periods to construct a valid counterfactual. Vessels appearing only after January 2024 (no pre-period baseline) or only before (no post-period reaction) are excluded. After this restriction, the final balanced sample comprises 15,781 unique vessels drawn from 19,887 active in the pre-period and 16,558 active in the post period, yielding a final panel of 1,755,789 vessel month observations.

The continuous outcome variables, namely *log_CO_2_per_nm*, *Avg Speed (kn)*, and *Distance (nm)*, are trimmed at the 1st and 99th percentiles to remove implausible sensor readings. Fuel share variables are bounded on $[0, 1]$ by construction and are not trimmed. A notable feature of the raw data is that 96% of HSFO consumption observations are missing. Given that the IMO 2020 global sulphur cap effectively phased out HSFO for vessels without scrubbers, these missing values reflect the absence of consumption rather than unrecorded data, and are therefore set to zero.

Each vessel is assigned a primary commodity label based on the type of cargo it most frequently carried across all its voyages in the dataset. This restriction matters because general cargo ships have much greater route flexibility than bulk carriers, they follow specific cargo wherever it needs to go, rather than running repeated loops between fixed loading and discharge ports. Including them risks attributing their naturally variable routing behavior to the EU ETS rather than to ordinary cargo logistics. Finally, vessels are assigned to a trade cluster based on the departure country of their voyages. Country-level means are computed for seven features (Distance (nm), *log_CO*2*_per_nm*, EU ETS exposure ratio, average speed, vessel DWT, MGO fuel share, and laden

ratio), standardized, and reduced via Principal Component Analysis (PCA). Ward hierarchical clustering is then applied to the resulting components, producing three country level profiles, based on 2023 voyage data exclusively, prior to the reform. While the silhouette score was marginally higher at four clusters, three was chosen because it produces a more interpretable economic story without meaningfully changing the substance of the results.

The algorithm assigns arbitrary numbers to each group, so the three clusters are then relabeled based on their average EU exposure: the group with the lowest average *ETS_Rate* becomes *Low EU*, the middle group becomes *Med EU*, and the most EU exposed group becomes *High EU*. The *Med EU* cluster should be seen as regrouping deep sea bulk fleet that passes through Europe on long hauls. Each vessel in the main panel then inherits the cluster label of its departure country; for instance, if France is assigned to *High EU*, a voyage departing from France is treated as *High EU* for that month. Because a vessel can depart from different countries over time, it is not locked into a single cluster; it is classified month by month according to where it sailed from.

A small number of countries did not fit cleanly into any of the three profiles and are grouped into a residual *Other* category, corresponding to 2,426 vessels in the balanced panel. Since this group has no coherent economic profile, it is excluded from the cluster-level regressions, though these vessels remain in the global regression sample.

3.3. Variable construction

***ETS_Rate*.** At the voyage level, *ETS_Rate* is mapped from the regulatory coverage rule: 1.0 for intra EU voyages, 0.5 for to/from EU voyages, and 0.0 for non EU voyages. After aggregation to the vessel month panel, it becomes a continuous mean in $[0, 1]$ reflecting the weighted average EU regulatory exposure across a vessel's voyages in a given month.

***Interaction* (DiD treatment variable).** $Interaction = ETS_Rate \times Post2024$. This is the variable whose coefficient is the central estimate of interest in every regression. It captures the differential change in outcomes associated with a one unit increase in regulatory exposure intensity after January 2024.

***ETS_x_Time* (trend control).** $ETS_x_Time = ETS_Rate \times TimeIdx$, where *TimeIdx* is a linear integer index of calendar months (January 2013 = 0, June 2025 = 149). This variable is included in the main specification (Model B) to absorb any pre existing linear behavioral trend proportional to EU exposure, and is discussed further in section 4.

***log_CO2_per_nm* (primary outcome).** The natural logarithm of total CO_2 emissions (tonnes) per nautical mile sailed. The log transformation ensures that regression coefficients are approximately interpretable as percentage changes and compresses the right tail of the raw distribution.

Fuel shares. *VLSFO_Share*, *HSFO_Share*, and *MGO_Share* are constructed as each fuel type's consumption divided by total fuel consumption. The three shares sum to 1 by construction.

3.4. Descriptive statistics and pre-reform trends

Table 1 reports summary statistics concerning the outcome variables used throughout the analysis for the full balanced panel of 1,755,789 vessel month observations. The median vessel month logs a CO_2 intensity of −1.54 ($\exp(-1.54) \approx 0.21$ tonnes of CO_2 per nautical mile), travels approximately 3,093 nautical miles, and steams at 11.15 knots. The fuel mix is dominated by VLSFO, which accounts for 90.3% of consumption on average, a direct consequence of the IMO 2020 global sulphur cap that effectively phased out HSFO for vessels without scrubbers. The right skew in HSFO share (mean 6.4% against a median of 0%) reflects its concentration among the minority of scrubber equipped vessels.

Table 1: Global summary statistics - Balanced panel (1,755,789 vessel-month observations)

Variable	Mean	SD	Min	P25	Median	P75	Max
$\log(CO_2/\text{nm})$	−1.561	0.471	−2.377	−1.966	−1.542	−1.233	−0.415
Avg Speed (kn)	10.97	1.71	6.40	9.89	11.15	12.18	14.57
Distance (nm)	4,136	3,795	214	1,319	3,093	5,478	19,028
MGO Share	0.033	0.047	0.000	0.008	0.016	0.038	0.905
HSFO Share	0.064	0.243	0.000	0.000	0.000	0.000	0.999
VLSFO Share	0.903	0.243	0.000	0.954	0.981	0.992	1.000

Table 2 reports raw mean changes in each of our outcome variable before and after January 2024. In this table, average speed fell by 4.6% and voyage distance contracted slightly, while CO_2 intensity per nautical mile rose by 2.6%. HSFO share increased by 24%, though this reflects a small base and is concentrated among scrubber equipped vessels rather than a more significant fleet shift. These raw comparisons do not isolate the policy effect from other shocks that happened during a similar time period, most notably the Red Sea crisis that occurred in the last months of 2023, but they confirm that the period around January 2024 does synchronize with some changes in the behavior of ships measured across most outcomes.

Table 2: Raw mean comparison before and after January 2024

Variable	Pre 2024 mean	Post 2024 mean	Change (%)
$\log(CO_2/\text{nm})$	−1.556	−1.597	+2.6
Avg speed (kn)	11.03	10.52	−4.6
Distance (nm)	4,145	4,070	−1.8
MGO share	0.033	0.034	+3.7
HSFO share	0.062	0.077	+24.1
VLSFO share	0.905	0.888	−1.8

Table 3 presents means broken down by trade cluster. The *High EU* cluster (13,664 vessels)

is characterized by short average voyages (2,805 nm), the highest MGO share (4.7%), and the lowest HSFO share (2.5%), consistent with short sea European trade where port emission regulations encourage cleaner fuels. The *Med EU* cluster (11,465 vessels) operates the longest routes (5,697 nm) and has the highest HSFO share (12.4%), suggesting a concentration of scrubber equipped bulk carriers on intercontinental hauls. The *Low EU* cluster (15,056 vessels) sits between these extremes, with moderate route distances and fuel shares.

Table 3: Variable means by trade cluster

Variable	**High EU** *(n=13,664)*	**Med EU** *(n=11,465)*	**Low EU** *(n=15,056)*	**Other** *(n=2,426)*
log(CO_2/nm)	−1.860	−1.234	−1.516	−1.951
Avg Speed (kn)	10.18	11.32	11.32	10.39
Distance (nm)	2,805	5,697	4,274	3,728
MGO Share	0.047	0.023	0.029	0.050
HSFO Share	0.025	0.124	0.062	0.012
VLSFO Share	0.928	0.853	0.909	0.939
Observations	533,273	383,878	829,533	9,105

The fuel mix variables reflect an important regulatory background. VLSFO (Very Low Sulphur Fuel Oil) became the dominant commercial fuel following the IMO 2020 global sulphur cap, which prohibited vessels without scrubbers from burning high sulphur fuels, and accounts for over 90% of consumption in this sample. HSFO (High Sulphur Fuel Oil) is the legacy heavy fuel, now legally confined to scrubber equipped vessels, with a median share of zero and a heavily right skewed distribution. MGO (Marine Gas Oil) is a cleaner, more expensive distillate fuel, which may explain its persistently higher average share among *High EU* vessels operating in European coastal waters.

We have decided to restrict the pre-trend analysis to 2023 rather than the full sample. The reason is simple: the long 2013-2025 window introduces structural breaks such as the IMO 2020 sulphur cap that make parallel trends harder to read visually. The year before the reform is the cleanest window to check whether the clusters were on comparable trajectories heading into January 2024. Figure 1 shows monthly cluster means for CO_2 intensity throughout 2023, the final pre-reform year. The three clusters exhibit flat, roughly parallel trajectories across all twelve months, providing visual support for the parallel trends assumption underlying the difference in differences design. Figure 2 shows the corresponding pattern for voyage distance, with the three clusters maintaining stable and parallel distance profiles throughout 2023, consistent with the parallel trends assumption for this outcome as well.

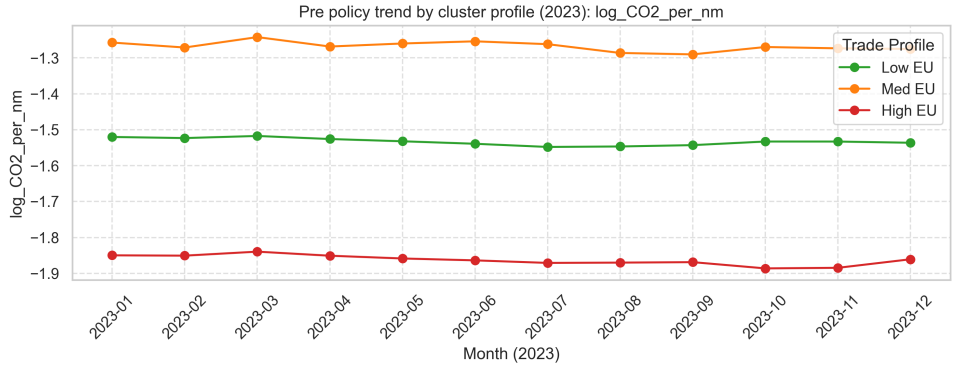


Figure 1: Pre-reform monthly means of log CO_2 intensity by cluster (January to December 2023)

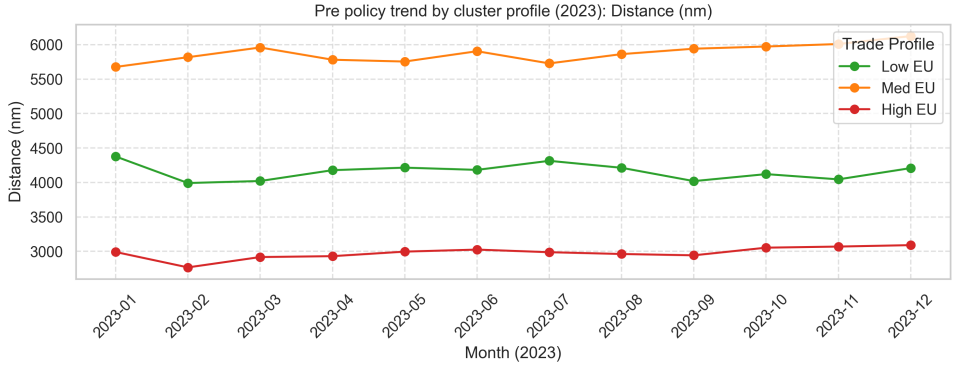


Figure 2: Pre-reform monthly means of voyage distance by cluster (January to December 2023)

Figure 3 shows the corresponding pattern for average speed. The three clusters maintain broadly parallel and flat trajectories throughout 2023, though a slight downward drift emerges toward the final months of the year for the *High EU* cluster, consistent with the early impact of the Red Sea rerouting shock. This motivates the inclusion of the trend control in Model B.

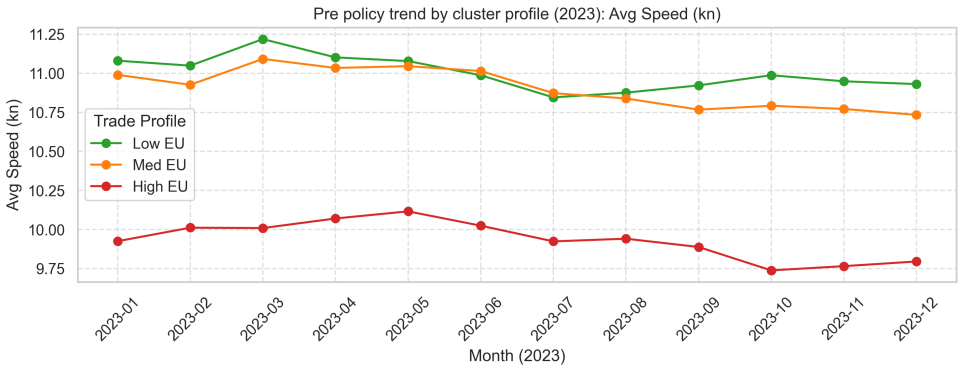


Figure 3: Pre-reform monthly means of average speed by cluster (January to December 2023)

Finally, figures 4, 5, and 6 show the patterns for fuel shares in the same calendar year. All clusters maintain stable and parallel profiles throughout 2023, providing some support for the parallel trends assumption for these outcomes.

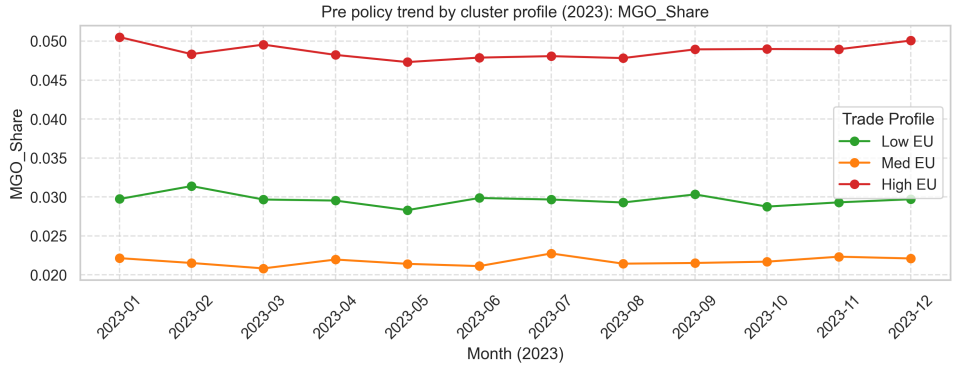


Figure 4: Pre-reform monthly means of MGO fuel share by cluster (January to December 2023)

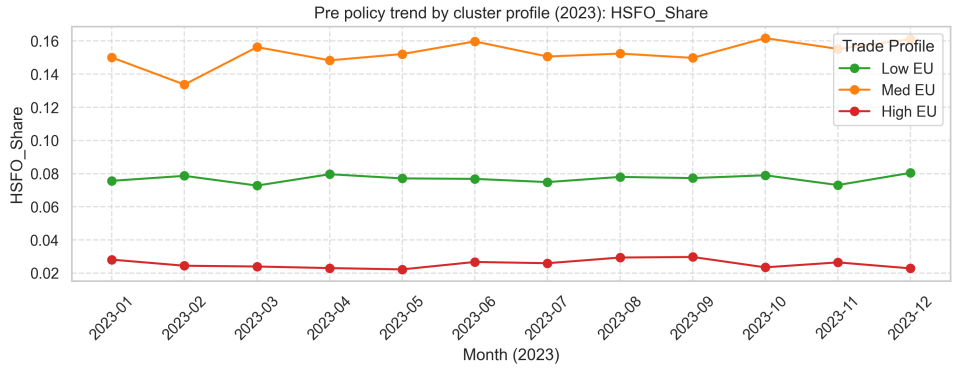


Figure 5: Pre-reform monthly means of HSFO fuel share by cluster (January to December 2023)

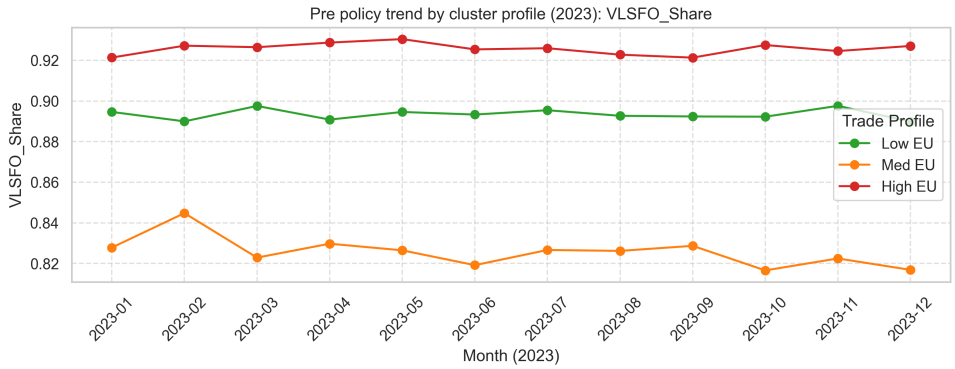


Figure 6: Pre-reform monthly means of VLSFO fuel share by cluster (January to December 2023)

4. Empirical strategy

4.1. Identification strategy

The core challenge of the analysis is related to the identification strategy. Indeed, the EU ETS took effect in January 1, 2024 for every vessel at the same time. Therefore, there were no group of ships that was unaffected by the policy. This rules out the standard binary DiD setup, where

you compare a treated group to an untreated one.

Instead of such a strategy, we exploit the fact that the policy does not hit all vessels equally hard. As described in section 1 Because the EU ETS applies 100% coverage on voyages within the EU but only 50% on voyages between an EEA and a non EEA port, vessels diverge in how much of their activity is actually taxed. Hence, a ship traveling only between several European ports does faces twice the cost of one that decided to split its time between European and non European routes. This variation in exposure, captured by *ETS_Rate*, is the pillar of our identification. We combine it with the January 2024 implementation date in a continuous treatment difference in differences design.

The identifying assumption is that, in the absence of the policy, vessels with different levels of EU exposure would have followed parallel trends over time, after accounting for vessel and month fixed effects. In other words, *High EU* and *Low EU* vessels should not have been diverging in speed, fuel use, or CO_2 intensity for reasons unrelated to the EU ETS heading into January 2024. The pre-trend figures in section 3.4 provide visual support for this on most outcomes. Knowing that, we have created a model where this assumption could be relaxed by allowing each vessel group to follow a pre-existing linear trend proportional to its EU exposure before the reform. Since all vessels face the same implementation date, the staggered treatment concerns raised by Callaway et al. (2024) and Sun and Abraham (2021) do not apply here.

4.2. The two specifications

We estimate two versions of the model for each outcome variable using a two way fixed effects estimator, which is simply a difference in differences with both vessel and time fixed effects included simultaneously. The comparison between the two versions is not just a robustness check; it is central to our argumentation.

4.2.1 Model A - Baseline DiD (no trend control)

$$Y_{it} = \beta_A \cdot Interaction_{it} + \delta \cdot ETS_Rate_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \tag{1}$$

where:

- Y_{it} is the outcome for vessel i in month t;
- $Interaction_{it} = ETS_Rate_{it} \times Post2024_t$ is the DiD treatment variable;
- ETS_Rate_{it} is the vessel's monthly mean EU coverage fraction;
- $Post2024_t$ is a dummy variable equal to 1 for $t \geq$ January 2024, else 0;
- α_i are vessel fixed effects, absorbing all time invariant vessel characteristics such as type, size, flag, or typical trade routes;

- γ_t are month fixed effects, absorbing any shock common to all vessels in a given month, including freight market cycles, the Red Sea disruption, IMO CII effects, and seasonality;
- ε_{it} is the error term.

The coefficient β_A is the key parameter of interests; it measures for a vessel moving from zero EU exposure to full EU coverage (one unit increase in ETS_Rate_{it}), how much does outcome Y change after January 2024 relative to before?

4.2.2 Model B - Main specification (with trend control)

$$Y_{it} = \beta_B \cdot Interaction_{it} + \delta \cdot ETS_Rate_{it} + \theta \cdot ETS_x_Time_{it} + \alpha_i + \gamma_t + \varepsilon_{it} \quad (2)$$

where $ETS_x_Time_{it} = ETS_Rate_{it} \times TimeIdx_t$, and $TimeIdx_t$ is a index of calendar months (January 2013 = 0, June 2025 = 149).

The added term in model B absorbs any incremental pre-existing trend in behavior that is proportional to EU exposure. If for some reasons *High EU* vessels were already slowly changing their speed or fuel mix before 2024, whether in anticipation of the policy or for other reasons, model A would inaccurately attribute that change to the shock of January 2024. Hence, the main specification (model B) forces $\beta_B$to capture only the shift at the reform date, after filtering for any pre-existing slope. If the coefficients estimates diverge significantly between model A and B in the result section (section 5), as they do most for average speed for instance, it would indicate that the simpler model was giving an inaccurate picture of the policy's true impact.

Both models are estimated using the *PanelOLS* estimator from the *linearmodels* Python library, with vessel and time fixed effects. Standard errors are clustered at the vessel level to account for vessel serial correlation across months, which makes sense because a vessel's behavior is likely correlated with its own past behavior.

4.3. Cluster level regressions

Beyond the global regressions on the entire panel, we estimate model B, the one with trend control, separately within each of the three trade clusters constructed using hierarchical clustering. To build these clusters, we computed averages at the country level of seven voyage characteristics (distance, CO_2 intensity, EU exposure, speed, vessel size, MGO share, and laden ratio) using 2023 data only. Afterwards, we ran a Principal Component Analysis (PCA) to compress them into a smaller set of components, and applied Ward's hierarchical clustering to produce three groups. Each ships then takes the cluster of its departure country, and the groups are relabeled by their average EU exposure as *Low EU*, *Med EU*, and *High EU*. For the DiD using the clusters, we kept the exact same structure as in the global specification (similar fixed effects and standard error clustering).

Table 4 is similar to table 3, but it regroups all the variables we used to build the clusters. The

numbers in it confirm that the groups are not arbitrary. Indeed, we notice that they are mapped onto three recognizable types of ship. The *High EU* cluster is made of small vessels, around 21,000 DWT, running short voyages of roughly 2,800 nm, which is the typical signature of short sea European trade. The *Med EU* cluster is the opposite: large ships of around 90,000 DWT on long trips close to 5,700 nm. Similarly, the *Low EU* cluster sits closer to the *Med EU* cluster than the *High EU* cluster. Furthermore, the fuel and speed variables are in accordance with this story, with the smaller *High EU* ships leaning more on the cleaner MGO.

Table 4: Cluster means on the variables used to build the clusters

Variable	**High EU** *(n=13,664)*	**Med EU** *(n=11,465)*	**Low EU** *(n=15,056)*	**Other** *(n=2,426)*
EU ETS exposure (*ETS_Rate*)	0.476	0.043	0.024	0.083
Distance (nm)	2,805	5,697	4,274	3,728
Avg Speed (kn)	10.18	11.32	11.32	10.39
$\log(CO_2/\text{nm})$	−1.860	−1.234	−1.516	−1.951
Vessel DWT	21,063	90,207	58,223	14,195
MGO Share	0.047	0.023	0.029	0.050
Observations	533,273	383,878	829,533	9,105

One might argue after reading this table that on EU exposure itself, the real contrast is between *High EU* and everyone else. As seen in this table; *High EU* vessels have almost half of their trade covered by the ETS, an exposure of 0.48, while both *Med EU* and *Low EU* sit close to zero, at 0.04 and 0.02. So the label *Med EU* should not be read as medium exposure. What separates the *Med EU* group from the *Low EU* one is mostly vessel size and route length, not how much they trade in Europe. We therefore read the clusters as three trade environments, where EU exposure is one dimension among several, rather than as a clean ladder running from low to high exposure. This is also why the cluster results in section 5 do not always line up well with the cluster names.

One of our design choice deserves to spend more time on it. As explained in the paragraph above, we cluster at the country level, not the vessel level. The reason is simple. A single vessel can have a noisy month: by that we mean that it might make an unusual detour, carry an atypical freight, or sit idle in port. If we clustered on individual vessels, those observations would give us messy, unreliable groups. Averaging at the country level fixes that. When many vessels depart from the same country, their idiosyncrasies cancel out. What remains is a stable picture of the trade environment associated with that country: its typical routes, its EU exposure, its fuel use, its vessel sizes. The objective of this is not to classify vessels. It is to classify the trade environments they operate in. Very importantly, the cluster label travels with the voyage, not with the ship. A vessel takes the label of the country it departed from in a given month. Obviously, since a single ship can load in different countries over the year, it can appear in more than one cluster

across our sample. This is why the count of the number of ships in table 3 represents more than the 15,781 vessels in the panel since the same vessel is counted once in each cluster it traded from, while the underlying vessel month observations are split cleanly across groups. We see this as a feature rather than a flaw. It means part of our identification comes from the same ship moving between trade environments, not only from comparing different ships. Within each cluster regression, a vessel's fixed effect is built only from the months it spent trading in that cluster.

Such a decomposition fulfill two purposes. First, it tests whether the change in behavior is stronger for vessels that faces higher regulatory exposure, which is what we would expect from a well functioning carbon price. Second, and more importantly, it prevents the model from masking opposed behavioral responses. Indeed, for instance, if some vessel groups increase CO_2 intensity while others decrease it, the global average coefficient may be close to zero not because there is no response, but because opposite effects neutralizes each other. As the results in section 5 show, this is exactly what is happening.

There is one design choice that deserves justification. Indeed, throughout our analysis we use three clusters, not two or four. The silhouette score was marginally higher at four clusters, but the fourth group did not add a distinct economic story. It split the medium EU ships into two groups with nearly identical behavior. Three clusters maps cleanly onto three types of commercial exposure: ships that almost never touch EU ports, vessels that do so occasionally, and vessels whose entire business relies on European trade. That distinction is economically meaningful, while four clusters is not.

4.4. Expected signs from theory

Before diving into the results, it is worth expanding on what standard theory would predict for each outcome.

Speed should fall. Slowing down reduces fuel consumption per nautical mile, and a higher carbon price makes that saving more valuable. Hence, CO_2 intensity per nautical mile should also decline if slowing down is the main strategy used to avoid paying higher tax.

Concerning the traveled distance, the mechanism is clear. For *High EU* vessels, rerouting cargo through a non- uropean hub shortens the taxed portion of the voyage and cuts the bill. Distance should fall. However concerning *Low EU* ships, we should observe the opposite behavior: they handled the freight diverted by the tax and sail longer routes. Which effect dominates at the aggregate level is what the data will tell us.

However fuel switching is harder to predict. MGO produces less CO_2 per unit of energy than VLSFO, but it is also more expensive, so switching is not obviously profitable just to save on emissions taxes. Whether it happens depends on how strictly EU port emission regulations are enforced. HSFO is legally restricted to scrubber equipped vessels and its use depends more on

the HSFO/VLSFO price differential than on the carbon price itself.

4.5. Threats to identification

The Red Sea shock is the most serious concern to our analysis. From late 2023, attacks on ships in the Red Sea forced a large share of the Asia/Europe trade to reroute around Africa via the Cape of Good Hope, adding thousands of nautical miles per voyage and affecting speeds and distances across the whole fleet. This shock is absorbed by the month fixed effects. The main threat is that the shock hit *High EU* and *Low EU* ships differently in proportion to their *ETS_Rate*. The trend control in model B addresses this partially. Indeed, the collapse of the speed coefficient from highly significant to close to 0 between model A and model B is consistent with the Red Sea crisis having created a different speed trend across vessels that model A was incorrectly reading as a policy effect.

The IMO 2020 CII regulation, which sets annual carbon intensity targets for all ships regardless of where they sail, is not a threat here. Since it applies to all vessels regardless of their route, its effect is fully absorbed by the month fixed effects and does not contaminate the estimates.

A sudden drop or surge in demand for a specific commodity in 2024 that happened to hit EU intensive routes harder than others would show up in our estimates as a policy effect even if it had nothing to do with the ETS. But persistent commodity characteristics are absorbed by the vessel fixed effects, since most bulk carriers specialize in a single cargo type. We did in our code an additional robustness check that further restricts the sample to pure bulk vessels and we obtained exactly the same results.

Additionally, the 30 month gap between the EU climate package announcement in July 2021 and the January 2024 implementation date means that some shipowners may have started adjusting their behavior before the official start date. If that is the case, then our DiD coefficient would capture only the response that had not already taken place, and all estimates should be read as conservative lower bounds on the full response to the policy. And that is precisely why a Regression Discontinuity Design (RDD) is not suited for this analysis.

One heterogeneity issue that we cannot fully tackle concerns laden versus ballast status. Indeed, empty vessels face less scheduling pressure and more freedom to slow down. If our results are driven by ballast legs, the significance would be smaller than the coefficients suggest. This could be an idea for future work with leg level data for instance.

Another way to push the current further is to replace the single *Post2024* dummy with a full set of time dummies, each interacted with *ETS_Rate*. This would create a standard event study, and could be seen as even more robust that the analysis done in this report. Instead of forcing the policy effect into one number, it lets us trace the effect period by period. The idea is simple. If our story is right, the interaction coefficients should be close to zero for every month before 2024, and only move once the policy starts, and can easily be observed in a graph. This gives us

two things at once. First, it is a visual check of parallel trends, without the pre-testing bias that Roth (2022) warns about, since we are not running a separate test and then deciding whether to trust the model. Second, it tells us whether the response is a clean break at January 2024 or a slow shift that began earlier. Concerning the latest, then this change would point to the Red Sea crisis rather than to the carbon price.

5. Main results

The following section will present the empirical results in two steps. The first part will discuss the choice of specification to show why the choice of model matters. The discussion will be focusing on the comparison of the baseline difference in difference of model A and the trend controlled main specification of model B. We then present the cluster level results, which are the core of the paper, organized around four empirical findings, namely (i) all cluster reduce speed post reform; (ii) the patterns of route differ across clusters consistently with carbon leakage; (iii) *High EU* vessels switch toward cleaner MGO and finally, (iv) CO_2 intensity responds asymmetrically across clusters, with most EU exposed vessels actually increasing their carbon intensity.

5.1. Why the choice of specification matters

Table 5: Global DiD results - Baseline vs. trend controlled specification

	Model A (baseline)		Model B (main spec)	
Outcome variable	Coef.	p-value	Coef.	p-value
log(CO_2/nm)	0.0167	0.000***	0.0039	0.020**
MGO share	0.0005	0.010***	−0.0001	0.667
HSFO share	−0.0002	0.106	0.0001	0.109
VLSFO share	−0.0004	0.126	0.0000	0.939
Avg speed (kn)	0.1796	0.000***	0.0102	0.448
Distance (nm)	286.55	0.000***	57.57	0.048**

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

Table 5 clearly seems to higlhy favor model B, compared to model A. Concerning the average speed outcome, the divergence between the two specifications is visible. Model A yields a coefficient of +0.1796 with a p-value of 0.000. However, once we add the pre-trend control in model B the coefficient shift to 0.0102 with a p-value of 0.448, effectively losing all significance that could be attributed to it in model A. This complete reversal of conclusion proves that the baseline DiD was not measuring a policy effect on speed at all, but rather a pre-existing differential trend. This difference in behavior could be associated with the Red Sea rerouting that began in late 2023.

A similar pattern is observed for MGO share where model A yields a coefficient of +0.0005

with a p-value of 0.010 while model B yields a coefficient of -0.0001 with a p-value of 0.667. While model A suggests a significant global increase in fuel use, model B suggests the opposite coefficient sign and a much lower significance. This divergence could wrongly lead us to believe that it is yet again a consequence of pre-existing trends, in which vessels were already moving towards MGO fuels before January 2024. However, as we shall prove further in the paper, the MGO effect is real but is concentrated in the *High EU* cluster. The aggregate result masks these contrasting dynamics by combining offsetting responses across clusters. Concerning the other fuel types, HSFO and VLSFO, neither model finds a significant result at the global level.

Two outcome survives the trend correction, CO_2 intensity and voyage distance in both specifications. However, we notice a 77% reduction in magnitude of the first one, with coefficient dropping from +0.0167** to +0.0039**, as well as in the second one, with coefficient dropping from +287*** to +58**. Both outcomes remain highly significant, giving us confidence that they reflect a genuine policy effect rather than a pre-trend artifact.

The lesson from table 5 is that the trend-controlled model B is not optional, and even necessary. If omitted, the speed result shows that a naïve DiD would have produced a large, significant yet sometimes economically counterintuitive finding. Furthermore, we could argue that the aggregate estimates alone are insufficient to characterize the effects of the EU ETS. The small but statistically significant global coefficients for CO_2 intensity (+0.004) and distance (+58 nm) mask a pronounced heterogeneity across vessel clusters. As we shall demonstrate in the next subsection, the substantive dynamics become apparent only through the cluster level analysis, to which we now turn.

5.2. The cluster results

Once we decompose the global fleet into three different clusters, we do obtain striking results worth commenting. Most of the outcome that appeared flat or insignificant at the global level in table 5 now shows significant effects, that are often in opposite directions across clusters, as we can see in table 6.

Table 6: Cluster level DiD results - Main specification (model B)

	Low EU		Med EU		High EU	
Outcome variable	Coef.	p-value	Coef.	p-value	Coef.	p-value
$\log(CO_2/\text{nm})$	−0.0227	0.015**	−0.0681	0.000***	0.0404	0.000***
MGO share	−0.0005	0.704	−0.0003	0.624	0.0037	0.000***
HSFO share	0.0001	0.824	−0.0012	0.034**	0.0000	0.200
VLSFO share	0.0005	0.734	0.0015	0.035**	−0.0037	0.000***
Avg speed (kn)	−0.1576	0.001***	−0.3039	0.000***	−0.1141	0.000***
Distance (nm)	772.96	0.000***	362.94	0.052*	−413.04	0.000***

$^{***}p < 0.01$, $^{**}p < 0.05$, $^{*}p < 0.10$.

5.2.1 Finding 1: speed reductions across all clusters, driven by different mechanisms

The first interesting results concerns the average speed outcome variable, where every single cluster reduced average speed after the reform, and every coefficient is highly significant. The *Med EU* cluster leads with a -0.3039*** kn reduction, followed by *Low EU* at -0.1576*** kn, and *High EU* at -0.1141*** kn.

The roughly 1.1% reduction in average cruising speed of the *High EU* cluster is the easiest to interpret. Indeed, it is consistent with the narrative of the behavioral change due to the EU ETS policy. The higher effective cost of fuel consumption imposed by the latter for EU covered voyages makes it profitable for fleets to slow down and burn less fuel per hour of operation. While the cost saving of a 1.1% decrease in speed is small in absolute terms, the counterfactual higher cost of the longer voyage duration seems to balance it at the current price level.

Concerning the *Med EU* cluster's speed reduction of 0.3039 knots is the sharpest of the three. It is partly explained by the higher flexibility they face. Indeed, even if it faces the same cost per tonne of CO_2 emitted constraints as the *High EU* in European waters, it has the ability to extend transit times without disrupting tight port scheduling due to their longer average voyage duration. This flexibility enables them to reduce speed more than the *High EU* cluster. Additionally, the *Med EU* cluster also faces the CII carbon intensity rating from the IMO. Their longer routes make slow pacing even more efficient regarding the reduction of their CII rating. The larger speed response can thus be explained by the combination of EU ETS compliance and CII rating pressure.

The *Low EU* cluster's speed reduction of 0.1576*** is the hardest result to explain. Indeed, these vessels have near zero EU exposure. Therefore they face almost no direct compliance cost. Two readings from those results are possible. The first is a general equilibrium effect: as *High EU* vessels rerouted, *Low EU* ships absorbed the diverted freight and faced busier ports with tighter scheduling. Slower approach speeds reduce congestion costs. The second is simpler and slightly more concerning: fuel prices were rising across the board in 2024, and slowing down is profitable regardless of carbon regulation. We cannot separate these two channels with our data, and that is a result worth studying further in an additional paper. What we can say is that the coefficient exists across all three clusters, which is more consistent with a fleet wide cost response than with a targeted compliance reaction.

Finally, as shown in figure 3, a small decline in average speed affected *High EU* cluster with vessels that traded mainly in European water in late 2023, coinciding with the Red Sea crisis. This common shock is fully absorbed in the main specification model used throughout the cluster analysis. The coefficients in table 6 identify only the differential speed response within each cluster, that is whether vessels with a higher ETS_Rate reduced speed more than vessels with a lower ETS_Rate in the same cluster. The speed results in table 6 therefore cannot be attributed to the Red Sea rerouting; they reflect vessel behaviour differences within clusters

that are correlated with EU exposure intensity.

5.2.2 Finding 2: route restructuring consistent with carbon leakage

The distance results in table 6 tells us another interesting story. It show that *High EU* vessels significantly reduced the length of their journey (-413.04***nm) while *Med EU* and *Low EU* vessels increased theirs, by 362.94 nm and 772.96*** nm respectively. This asymmetric pattern is consistent with carbon leakage through cargo and route diversion, this is the carbon leakage effect in action.

The *High EU* speed contraction could be explained from an economical point of view. Knowing that the pre-reform mean voyage of the *High EU* cluster was approximately 2,800 nautical miles, the 413 nm contraction represents a roughly 15% reduction in voyage length for the most exposed vessels. Since the EU ETS charges shipowners per tonne of CO_2 emitted during a voyage, a shorter route directly translates into fewer permits to request. For a vessel making frequent intra EU stops, even a small reduction in distance can therefore generate important savings over the course of a year. Shortening voyage distance could therefore be seen as a rational behavior to the carbon charge, one that requires no capital investment in the long term, nor contract renegotiation, and can be implemented immediately by simply rerouting through a non EEA intermediate port.

The expansion of 772.96*** nm of the *Low EU* cluster is the clearest empirical signature of carbon leakage in the data. While *High EU* vessels were affected by higher compliance costs, the demand for *Low EU* vessels increased, effectively expanding voyage distance by around 21% for the most affected vessels within the cluster. That disturbance hit vessels that navigated on the Asia/Europe corridor, which is not where *Low EU* ships predominantly operate. The fact that it is precisely the least regulated ships that expanded their routes after January 2024 points more naturally to cargo diversion than to any broader global shipping shock.

The *Med EU* increase in voyage distance of 362.94* nm points in the same direction. It is also consistent with the divergence of cargo form the *High EU* operators. The marginal significance of this result reflects greater heterogeneity within this cluster. Some *Med EU* vessels suffer more from the EU ETS policy reform than others, who expand similarly to the *Low EU* cluster. This hand in hand trend between some *Med EU* and *Low EU* vessels proves that the carbon leakage effect is not only confined to the most non EU vessels.

The results discussed above show sobering implications for the environmental effectiveness of the EU ETS reform. If *High EU* vessels contract routes distance by 413.04*** nm while *Low EU* ones expand it by 772.96*** nm, and if *Low EU* vessels are more numerous in absolute terms than *High EU* ones, the net effect on total nm sailed across the globe is likely positive. Indeed, longer distances mean more fuel burned and potentially more total CO_2 emitted globally. Rather than lowering global shipping emissions, the EU ETS reform may well simply redistribute it.

5.2.3 Finding 3: *High EU* vessels switch to cleaner fuel : the policy working as intended

Here the policy delivers a clean and positive result. We observe a statistically significant shift in fuel mix post-reform of the *High EU* vessels of 0.0037*** in MGO share and of -0.0037*** in VLSFO share. Meanwhile, no significant shift in fuel mix is observed in the *Low EU* nor *Med EU* clusters. These findings are exactly as we would expect from an efficient policy: there is an incentive to switch to cleaner fuel, especially in the most EU exposed part of the fleet.

The mechanism is rather intuitive. Under carbon pricing, the slightly lower CO_2 content of MGO relative to VLSFO makes it marginally more attractive for the regulated fleet. Additionally, the fact that MGO is already required in most EU ports in the EU Emission Control Areas means that *High EU* vessels are already equipped with the infrastructure and regulatory context for MGO use. The EU ETS regulation only deepens the cost benefit calculus towards MGO use on the EU covered legs by enforcing a compensation by extra permits for every tonne of VLSFO burnt. This intuition is confirmed by the insignificant impact on the *Low EU* and *Med EU* clusters. Taken together, these results suggest that the policy is working as intended for this dimension: the most regulated vessels are gradually shifting toward cleaner fuel, and the effect is concentrated exactly where the carbon price hits the hardest.

The magnitude is modest, of 0.37%, and it can be easily explained by two factors. The first one is the highly concentrated fuel shares in VLSFO post-IMO 2020. *High EU* vessels use approximately 92 to 93% VLSFO pre-reform, leaving a small room for switching. Secondly, the effective 24 euros per tonne carbon cost in 2024 only produces a modest cost differential between VLSFO and MGO. From 2026 onward, the full permit obligation, raising the effective carbon cost to 60 euros per tonne makes the incentive to switch to MGO roughly 2.5 times stronger. We should expect to see this switch effect to grow substantially.

One additional and rather unexpected finding emerges from the *Med EU* cluster. In the latter, we observe a decrease in HSFO share by 0.0012** while VLSFO share increases by 0.0015**. Pre-reform, the *Med EU* cluster had higher shares compared to the other clusters. It is thus very likely that they retained some HSFO use post-IMO 2020. The EU ETS further incentivizes the retirement of HSFO in favor of VLSFO due to the higher carbon content of HSFO. This results suggest that the fuel switching from HSFO to VLSFO is occurring in the one that had the most switching left to do.

5.2.4 Finding 4: the CO_2 intensity paradox

The most interesting findings in this paper goes directly against the stated purpose of the EU ETS. Indeed, we found that *High EU* vessels, the most heavily regulated group, display a statistically significant 0.0404*** increase in log CO_2 per nautical miles while *Med EU* and *Low EU* vessels both display statistically significant decreases of 0.0681 and 0.0227 respectively. These results directly contradict the aim of the EU ETS policy as the vessels being regulated the most are

becoming the least efficient.

This positive *High EU* coefficient has a coherent economic explanation, directly tied to the second finding. *High EU* vessels contracted their route distance by 413.04*** in response to the EU ETS reform. The important twist to note here is that shorter distances also imply more manoeuvres when approaching and departing ports, which other clusters do not suffer from due to their less effective numerous stops over their longer voyages. The latter mechanism implies more CO_2 emissions for *High EU* vessels. The route contraction thus constrains *High EU* vessels to more fuel inefficient and port intensive operational profiles. The metric taxed (total covered emissions) is moving in the opposite direction as the metric that ultimately matters for the climate (global CO_2 per unit of transport work). The *Med EU* cluster does illustrate this contrast perfectly. Its CO_2 intensity decreased by 0.0681***, which is the largest decline of the three while it simultaneously reduces speed by 0.3039*** knots and expands voyage distance by 362.94* nm. These findings correspond to the textbook slow steaming response. When maintaining its long distance routes but reducing its speed, CO_2 per nautical miles traveled by the *Med EU* cluster drops. *High EU* vessels chose a different strategy, they contracted their routes rather than decreased their speed. What we observe is thus two qualitatively different compliance strategies with opposite effects on CO_2 per nm.

The *Low EU* results complete the pictures. Those vessels traveled 772.96*** nm more while decreasing their CO_2 per nm by 0.0227**. To understand this apparent contradiction, we need to observe the nature of the additional routes absorbed by the *Low EU* cluster. If the diverted cargo flows onto very long haul routes, the latter are characterized by extended periods of high efficiency open ocean cruising with minimal port cost fractions. These long distance routes are structurally more CO_2 efficient than shorter ones. So even as these vessels travel further and emit more in absolute terms, their efficiency per nautical mile improves because the proportion of their time spent in port is lower.

Putting it together : *High EU* vessels increase CO_2 per nm by around 4% on a shorter route distance while *Low EU* decrease it by around 2.3% on a longer route. Considering the *Low EU* distance increase to be roughly twice the magnitude of the *High EU* distance decrease and assuming comparable vessel sizes and emissions rates, the *Low EU* expansion in total voyage emissions likely exceeds the *High EU* reduction. The EU ETS, in its first year, may well have produced a net increase in global shipping CO_2. Confirming this quantitatively would require total fleet emissions estimates beyond what DiD coefficients alone can provide, and we flag this as an important priority for future work.

6. Policy implications

The results of section 5 describes a mixed picture of a policy that is beginning to impact ships behaviors, but with some unexpected consequences. From our empirical findings, we can draw

three implications.

6.1. The 50/100 coverage rule is generating the leakage it was supposed to avoid

The most striking result in the paper is the distance divergence. *High EU* vessels shortened their average monthly voyage distance by around 413 nautical miles after January 2024, while *Low EU* vessels followed the opposite trajectory, and expanded theirs by 773 nautical miles over the same period.

The 50/100 coverage rule described in section 1 is the key driver of this phenomena. A ship on a route within the EEA must cover the full carbon cost of its emissions, while a vessel arriving from outside the EEA only pays for half. This creates a clear incentive: by deviating through a non EEA port before the final EEA destination, rather than directly from the origin port, a shipowner would reduce its operating cost by a significant margin. The fact that *Low EU* vessels expanded their routes by nearly twice as much as *High EU* vessels contracted theirs is consistent with this exact mechanism, and suggests that cargo is being rerouted rather than a drop in the flow of cargo.

The problem is that rerouting does not reduce global emissions. What changes is not how much a given freight moves, but how many vessels are needed to transport it. A direct voyage to a European port is replaced by two shorter trips, one to a non EEA hub, and one for the final leg into Europe. Each short trip carries a fixed emissions cost from port maneuvering and departure that a single long voyage would not. Our CO_2 intensity results are consistent with this: *High EU* vessels, which are doing more of these shorter, port intensive trips, actually became less efficient per nautical mile after the reform.

The most direct fix would be to move toward 100% coverage on all voyages with at least one EEA port call, regardless of where the voyage originated. This would eliminate this cost asymmetry that makes these detours financially attractive. However, it is worth noting that the political obstacle is real: non EU countries have consistently pushed back against the EU applying its environmental rules beyond its own borders. The 2012 attempt to include international aviation in the ETS is a perfect example: it was suspended within a year following strong objections from the US, China, and Russia. Extending the policy to cover all emissions on voyages arriving from outside the EEA would likely face similar resistance under UNCLOS and WTO frameworks. But the alternative is a policy whose diversion incentives will only grow stronger now that the permit obligation has reached 100% in 2026, since a higher carbon price makes rerouting proportionally more profitable (Dominioni et al., 2018; Wu et al., 2024).

6.2. A differentiated obligation by trade intensity

The three cluster structure that resulted from our PCA is not a simple methodological choice. We believe that it reveals something real about how the fleet operates. Indeed, we could hardly argue that a vessel making weekly port stops between Hamburg and Rotterdam for instance has a completely different relationship to the EU ETS than one that stops at a single European port twice a year. However, the policy hurt the ships similarly, and our results in section 5 clearly show that there answer to the tax is also different.

Such a phenomena points to a policy design that has not, to our knowledge, been proposed for shipping in the EU: a regulation that would charge shipowners more if they trade more in European waters. The idea is simple; vessels that trade on a weekly basis in the EEA would face a higher coverage rate; vessels that only occasionally stop at European ports would face a lower one. The total amount of emissions covered would not change, only how the burden is shared.

This policy design is not as new as it sounds. The EU ETS already does something close to this for heavy industry. Under phase 3, sectors such as steel and cement that were both carbon intensive and heavily exposed to foreign competition received more free allowances than sectors that were not, precisely because they faced a real risk of moving production outside the EU to avoid the cost (Martin et al., 2014). The logic of this policy is the same as the one we are proposing in this subsection; to adjust the burden to reflect actual exposure, rather than applying the same rule to every kind of ships, without even considering their actual trade intensity in EU waters.

The logic works differently here than it did for industry when the EU ETS was implemented for this sector. In phase 3, free permits went to firms most at risk of leaving the EU. Concerning shipping, the *High EU* vessels are the least likely to reroute, because Europe is their core market. The vessels that reroute are in the Low and Medium EU clusters. So the differentiated tax should work the other way around: higher tax pressure on the ships that rarely enter EU waters, who have the most to gain from diversion, and a more stable regime for the other kind of ships, who are already locked into EU trade and cannot easily escape anyway. This does not reduce the environmental ambition of the policy, it just places the burden where the behavioral response is smallest.

That would have a dual benefit. For *High EU* vessels, a smaller obligation removes the incentive to dodge European ports, since the cost of rerouting would exceed the saving from the carbon tax. For *Low EU* vessels, that currently heavily benefit from that rerouting, a higher tax would reduce the financial gain from that diversion, weakening the other end of the problem at the same time.

6.3. Rethinking what the tax actually measures

According to us, the most original finding in this paper, is the CO_2 intensity paradox documented in the fourth finding in the section 5. In it, we observe that, for the cluster with the highest trade

intensity toward the EU which are the ships targeted by the policy, became less CO_2 efficient per nautical mile after the reform. Paired with the fact that this cluster group shortened their trips, more frequent port stops mean that maneuvering, slowing down, waiting, and speeding back up burn far more fuel per mile than cruising in open water with less stops.

Actually, that draw attention into a deeper issue with how the policy defines what it is actually trying to minimize. Indeed, the criterion used to observe the effectiveness of the the policy is the total emissions on voyages reaching EU ports. But the metric that lawmakers ultimately care about is closer to CO_2 per unit of transport work performed, that is, how much carbon is emitted to move a given amount of freight in a given distance. These two metrics do not always move together, and our results suggest that the current design can improve one while worsening the other.

What our results add to this debate is a empirical illustration of why the current approach can backfire. Taxing only the European portion of a voyage creates an incentive to shrink that portion, which, as we demonstrate, raises the carbon cost per unit of freight moved rather than lowering it. Fixing the 50% rule, as argued in the previous subsection, would address part of this issue. But as long as the policy tracks voyage level emissions rather than transport efficiency, the misalignment between what the tax measures and what it is trying to achieve will remain.

7. Conclusion

This paper asked whether the January 2024 extension of the EU ETS to maritime shipping changed how ships behave. Using a large panel of bulk vessels observed over more than a decade, we tested four outcomes: sailing speed, route distance, fuel use, and CO_2 intensity.

The answer across all four dimensions is mixed. On speed, what initially appeared as a policy response turned out to reflect a pre-existing differential trend linked to the Red Sea crisis. Once we accounted for it, no meaningful slowdown could be attributable to the carbon price. The clearest and most robust finding is on routes: vessels heavily exposed to the regulation pulled back from longer European voyages while those operating mostly outside EU waters expanded theirs, a pattern that points directly to cargo being diverted away from European ports rather than any genuine reduction in shipping activity. A moderate shift toward cleaner fuel is visible among the most regulated vessels, but again, the results remain small which makes sense as the effective carbon cost was in the first compliance year. The most striking result, however, runs counter to the policy's own objective. The vessels bearing the highest regulatory burden, against all odds, became less efficient. By shortening their routes to reduce compliance costs, they ended up making more frequent port stops, and the emissions associated with each port call are spread over far fewer nautical miles than on the longer voyages they replaced.

What emerges from these findings is a policy that is beginning to have measurable effects, but not where it counts. Operators are responding to the carbon price by adjusting routes, trimming

speeds and adjusting fuel choices, yet the net effect appears to be a reshuffling of emissions across the globe rather than a reduction in them. When regulated vessels contract their European routes, the cargo does not disappear; it moves onto longer, unregulated voyages handled by vessels with little to no EU exposure. For a policy whose stated purpose is to reduce shipping's contribution to global warming, this is a fundamental problem, one that stems directly from trying to apply a regional instrument to an industry that is, by nature, global.

Three limitations should be kept in mind. First, the analysis covers only the first compliance year, operators were required to cover only 40% of their emissions. We assume that behavioral responses should grow further as the obligation reaches 100% in 2026. Second, the Red Sea crisis remains an imperfectly resolved confound, particularly for the speed and CO_2 intensity results. Third, our empirical model identifies how different vessel groups responded relative to one another, but it does not directly measure what happened to total global shipping emissions as a result.

We encourage future work to exploit post-2026 data and apply a similar framework to it, as the move to full compliance will provide a much sharper test of whether the behavioral responses documented here scale with the carbon price as theory predicts. Beyond that, the broader conclusion this paper reaches is one environmental economics has arrived at before: for a globally mobile industry emitting a global pollutant, regional carbon pricing alone is not enough. Closing the leakage channel requires either reforming the EU ETS coverage rules or advancing a coordinated global carbon levy at the IMO.

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